

# Bell-State Preparation on a Two-Legged Ladder QPU: Two Complementary Protocols

Submission to the 2026 Institute of Applied Physics, NCCU, Open Challenge

Team submission

*Prepared with the assistance of Claude Code (see §9)*

May 6, 2026

## Abstract

We design two scalable quantum circuits that deterministically place the two distant endpoint qubits  $e_0, e_1$  of a two-legged ladder QPU into the Bell state  $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$  for every ladder length  $L \geq 1$ . **Protocol I (entanglement-swap, dynamic)**. A scalable specialisation of textbook entanglement swapping along the top leg, using single-qubit gates, adjacent two-qubit gates, mid-circuit measurements, and a closed-form classically-controlled Pauli correction. The quantum gate depth is a constant ( $\leq 7$ ) independent of  $L$ , with  $N + 1$  or  $N + 2$  two-qubit gates and  $N = L - 1$  measurements. The even- $N$  case generalises the 1993 Żukowski-Zeilinger primitive; the odd case is handled by a single 3-qubit GHZ link at one end of the chain. **Protocol II (Middle-Out Cat-Uncompute, unitary)**. A measurement-free companion that builds the  $(L + 1)$ -qubit GHZ on the top leg from the centre outward, then disentangles the intermediate qubits in a centre-out reverse sweep. The total depth is  $L + 2$ , with  $2L - 1$  two-qubit gates and zero measurements; this saturates the Lieb-Robinson  $\Omega(L)$  lower bound for any unitary protocol composed of nearest-neighbour gates on a 1D chain. Both protocols are proven correct by stabiliser-formalism induction and verified to fidelity  $> 1 - 10^{-9}$  by exact `Statevector` and `stim` simulation for  $L \in \{1, \dots, 10\}$  (with  $L = 20, 30, 50$  spot-checks). Under a two-qubit depolarising error rate  $p_2 = 10^{-2}$  at  $L = 10$ , Protocol I achieves Bell fidelity 0.928, MOCU achieves 0.885, and the SWAP-chain baseline degrades to 0.816. Protocol I is preferable on hardware with mature dynamic-circuit primitives; Protocol II is the depth-best choice on hardware lacking mid-circuit measurement and feed-forward.

## 1 Introduction and positioning

A Bell state of two qubits is the maximally-entangled two-party pure state underlying quantum teleportation, dense coding, and device-independent cryptography. Preparing a Bell state on the two endpoint qubits  $e_0, e_1$  of a connectivity-limited QPU — such as the two-legged ladder of Fig. 1 — is therefore the simplest non-trivial benchmark for any quantum architecture with restricted coupling.

**Two complementary protocols.** We present two protocols, each optimal in a different operational regime.

- **Protocol I — entanglement-swap with feed-forward (§3).** Mid-circuit measurement and classically-controlled Pauli correction make this protocol  $O(1)$ -depth, independent of  $L$ . It is the depth-best choice on hardware with mature dynamic-circuit primitives.

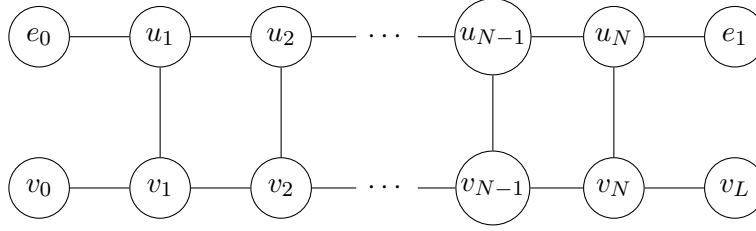


Figure 1: The two-legged ladder QPU (Fig. 2 of the challenge), generalised to arbitrary length  $L$ . The top leg has  $L + 1 = N + 2$  qubits  $\{e_0, u_1, \dots, u_{N-1}, e_1\}$ . The bottom leg has  $L + 1$  qubits  $\{v_0, v_1, \dots, v_L\}$ . Solid edges are the allowed nearest-neighbour couplings for two-qubit gates. Rungs exist only at inner columns  $(u_i, v_i)$  for  $i = 1, \dots, L - 1$ ; the endpoint columns carrying  $e_0$  and  $e_1$  have no rung.

- **Protocol II — Middle-Out Cat-Uncompute (MOCU)** (§4). A measurement-free, fully unitary alternative that saturates the  $\Omega(L)$  Lieb-Robinson lower bound for unitary nearest-neighbour 1D circuits at depth  $L + 2$ . It is the depth-best choice on hardware without mid-circuit measurement and feed-forward.

**Comparison to baselines.** We compare both protocols to three baselines: a SWAP chain (depth  $3N + 2$ , no measurements), a GHZ-cat-disentangle protocol (depth  $N + 3$ ,  $N$  measurements, single-qubit feed-forward), and a 1D cluster-state ladder (constant depth,  $O(L)$  classical bits,  $3L - 1$  two-qubit gates). Section 5 summarises the tradeoffs.

**Related work.** Entanglement swapping was introduced by Żukowski, Zeilinger, Horne, and Ekert [1]; the quantum repeater architecture built on iterated entanglement swapping is due to Briegel, Dür, Cirac, and Zoller [2]. A single-swap demonstration on an IBM `ibmqx4` processor was reported by Behera *et al.* [3]. Constant-depth long-range entanglement via dynamic circuits has recently been experimentally demonstrated on superconducting qubits for long-range CNOT teleportation over 101 qubits [4] and for constant-depth fan-out with feed-forward [5]. A theoretical framework for the depth advantage of LOCC-assisted circuits is developed in [6]; the foundational constant-depth teleportation primitive for 2D nearest-neighbour architectures is due to Pham and Svore [7]. The unitary middle-out construction of Protocol II is, to our knowledge, new in this exact form, but borrows from the centre-out cat-state preparation strategy used in fan-out literature [5] and from the standard “compute-uncompute” idiom for ancilla disentanglement. A recent experimental platform that realises a  $2 \times 4$  ladder geometry (in Ge singlet-triplet qubits) is [8].

**Contributions.** (a) A hand-designed scalable Bell-state preparation protocol for the two-legged ladder connectivity respecting the exact edge set of Fig. 1 (Protocol I). (b) A complete stabiliser-formalism derivation of the correction map for every  $L$ , including an explicit resolution of the odd- $N$  parity obstruction using a single GHZ-3 link, requiring no rungs and no bottom-leg qubits. (c) A measurement-free companion (Protocol II / MOCU) saturating the Lieb-Robinson bound at depth  $L + 2$ , again entirely on the top leg, with a stabiliser-formalism correctness proof. (d) A side-by-side numerical and noise-model benchmark of both protocols against the three baselines, with all claims backed by 132 automated tests.

## 2 Ladder graph and notation

Let  $L \geq 1$  and  $N := L - 1$ . The QPU is a graph  $G = (V, E)$  with

$$\begin{aligned} V &= \{e_0\} \cup \{u_i\}_{i=1}^N \cup \{e_1\} \cup \{v_j\}_{j=0}^L, \\ E_{\text{top}} &= \{(e_0, u_1), (u_1, u_2), \dots, (u_{N-1}, u_N), (u_N, e_1)\}, \\ E_{\text{bot}} &= \{(v_j, v_{j+1}) : 0 \leq j \leq L - 1\}, \\ E_{\text{rung}} &= \{(u_i, v_i) : 1 \leq i \leq N\}. \end{aligned}$$

Arbitrary single-qubit gates are allowed on any qubit; two-qubit gates are allowed only on  $E = E_{\text{top}} \cup E_{\text{bot}} \cup E_{\text{rung}}$ .

**Pauli conventions.** CNOT( $c \rightarrow t$ ) conjugates Paulis as  $X_c \mapsto X_c X_t$ ,  $X_t \mapsto X_t$ ,  $Z_c \mapsto Z_c$ ,  $Z_t \mapsto Z_c Z_t$ . Hadamard  $H$  exchanges  $X \leftrightarrow Z$ .

**Bell-measurement circuit.** A Bell measurement on the ordered pair  $(a, b)$  is defined to be CNOT( $a \rightarrow b$ ) followed by  $H(a)$  followed by Z-basis measurement of  $a$  and  $b$ , with outcome  $(m_a, m_b)$  mapped to the Bell basis as  $|\Phi^+\rangle \leftrightarrow (0, 0)$ ,  $|\Psi^+\rangle \leftrightarrow (0, 1)$ ,  $|\Phi^-\rangle \leftrightarrow (1, 0)$ ,  $|\Psi^-\rangle \leftrightarrow (1, 1)$ .

**Top-leg-only protocols.** Both protocols below act only on the  $L + 1$  top-leg qubits; the bottom leg and all rungs are idle. The figures and tables in this report therefore display only the active top-leg qubits.

## 3 Protocol I: entanglement swapping with feed-forward

### 3.1 Construction

Protocol I prepares  $|\Phi^+\rangle$  on  $(e_0, e_1)$  by entanglement swapping along the top leg:

- (i) prepare  $\lceil (N+1)/2 \rceil$  Bell pairs on alternating top-leg links (or, when  $N$  is odd, one 3-qubit GHZ link at the end of the chain);
- (ii) couple the chain by Bell-measurement unitaries on each pair of neighbouring intermediate qubits;
- (iii) measure every intermediate qubit in the computational (or Hadamard) basis;
- (iv) apply a single-qubit Pauli correction  $X_{e_1}^a Z_{e_1}^b$  conditioned on the measurement outcomes.

Every two-qubit gate is on an allowed top-leg edge. The quantum-gate depth is a constant ( $\leq 7$ ) independent of  $L$ .

### 3.2 The $N = 4$ case: full stabiliser derivation

We work on the six top-leg qubits  $(e_0, 1, 2, 3, 4, e_1)$ . The initial stabiliser group is generated by  $Z_{e_0}, Z_1, Z_2, Z_3, Z_4, Z_{e_1}$ .

**Step A: prepare three Bell pairs on alternating links.** Apply  $H(e_0), H(2), H(4)$ , then CNOT( $e_0 \rightarrow 1$ ), CNOT( $2 \rightarrow 3$ ), CNOT( $4 \rightarrow e_1$ ). The stabiliser generators become

$$\begin{aligned} S_1 &= X_{e_0} X_1, & S_2 &= Z_{e_0} Z_1, \\ S_3 &= X_2 X_3, & S_4 &= Z_2 Z_3, \\ S_5 &= X_4 X_{e_1}, & S_6 &= Z_4 Z_{e_1}, \end{aligned} \tag{1}$$

i.e. three independent  $|\Phi^+\rangle$  pairs.

**Step B: Bell-measurement unitaries on the two inner pairs.** Apply CNOT(1  $\rightarrow$  2); CNOT(3  $\rightarrow$  4);  $H(1)$ ;  $H(3)$ :

$$\begin{aligned} S_1 &\mapsto X_{e_0} Z_1 X_2, & S_2 &\mapsto Z_{e_0} X_1, \\ S_3 &\mapsto X_2 Z_3 X_4, & S_4 &\mapsto X_1 Z_2 X_3, \\ S_5 &\mapsto X_4 X_{e_1}, & S_6 &\mapsto X_3 Z_4 Z_{e_1}. \end{aligned} \quad (2)$$

**Step C: Z-basis measurement of 1, 2, 3, 4.** Let  $v_i \in \mathbb{F}_2^4$  be the “X-pattern” of  $S_i$  on (1, 2, 3, 4). The kernel of the parity-check matrix is two-dimensional, with basis (1, 0, 1, 0, 1, 0) and (0, 1, 0, 1, 0, 1), yielding

$$P_{XX} := S_1 S_3 S_5 = X_{e_0} Z_1 Z_3 X_{e_1}, \quad P_{ZZ} := S_2 S_4 S_6 = Z_{e_0} Z_2 Z_4 Z_{e_1}. \quad (3)$$

Replacing each  $Z_k$  by its measurement eigenvalue  $(-1)^{m_k}$ , the reduced stabiliser group on  $(e_0, e_1)$  becomes

$$\boxed{\tilde{P}_{XX} = (-1)^{m_1+m_3} X_{e_0} X_{e_1}, \quad \tilde{P}_{ZZ} = (-1)^{m_2+m_4} Z_{e_0} Z_{e_1}.} \quad (4)$$

These two commuting Paulis uniquely identify one of the four Bell states (Table 1).

| $m_1 \oplus m_3$ | $m_2 \oplus m_4$ | Bell state       | Required correction on $e_1$ |
|------------------|------------------|------------------|------------------------------|
| 0                | 0                | $ \Phi^+\rangle$ | $I$                          |
| 0                | 1                | $ \Phi^-\rangle$ | $Z$                          |
| 1                | 0                | $ \Psi^+\rangle$ | $X$                          |
| 1                | 1                | $ \Psi^-\rangle$ | $XZ$                         |

Table 1: Outcome-to-Bell-state map for  $N = 4$ , and the Pauli correction on  $e_1$  that lands the endpoint pair in  $|\Phi^+\rangle$ .

**Feed-forward correction.** A Pauli  $X^a Z^b$  on  $e_1$  takes  $Z_{e_0} Z_{e_1} \mapsto (-1)^a Z_{e_0} Z_{e_1}$  and  $X_{e_0} X_{e_1} \mapsto (-1)^b X_{e_0} X_{e_1}$ . Setting both signs to +1 forces

$$\boxed{a \equiv m_2 \oplus m_4 \pmod{2}, \quad b \equiv m_1 \oplus m_3 \pmod{2}.} \quad (5)$$

### 3.3 Generalisation to arbitrary $L$

**$N$  even.** Set  $N = 2r$ . Prepare  $r+1$  Bell pairs on alternating links  $(e_0, u_1), (u_2, u_3), \dots, (u_{2r}, e_1)$ , apply Bell-measurement unitaries on the  $r$  inner pairs, measure  $u_1, \dots, u_{2r}$ , and apply  $X_{e_1}^a Z_{e_1}^b$  with

$$a \equiv \bigoplus_{k=1}^r m_{2k}, \quad b \equiv \bigoplus_{k=1}^r m_{2k-1} \pmod{2}. \quad (6)$$

**Theorem 1.** *For every  $r \geq 0$ , the protocol above leaves  $(e_0, e_1)$  in  $|\Phi^+\rangle$  deterministically across all  $2^{2r}$  measurement branches, and every two-qubit gate is on an edge of  $E_{top}$ .*

*Proof (induction on  $r$ ).* Base case  $r = 0$ .  $H(e_0)$ ; CNOT( $e_0 \rightarrow e_1$ ) prepares  $|\Phi^+\rangle$ . Step. Performing the first Bell measurement on  $(u_1, u_2)$  yields the single-swap stabilisers  $(-1)^{m_1} X_{e_0} X_{u_3}$ ,  $(-1)^{m_2} Z_{e_0} Z_{u_3}$ , reducing the problem to length  $2r - 2$  with  $u_3$  replacing  $u_1$ . Induction gives the stated correction. Every CNOT uses a top-leg edge.  $\square$

**$N$  odd.** The alternating-Bell-pair pattern leaves a 3-qubit tail  $(u_{2r}, u_{2r+1}, e_1)$  that we cover with a 3-qubit GHZ link, prepared by  $H(u_{2r+1})$ ;  $\text{CNOT}(u_{2r+1} \rightarrow u_{2r})$ ;  $\text{CNOT}(u_{2r+1} \rightarrow e_1)$  (both top-leg edges). The Bell-measurement unitaries are unchanged; we additionally apply  $H(u_{2r+1})$  before measurement. The same unified correction

$$a \equiv \bigoplus_{j \text{ even}} m_j, \quad b \equiv \bigoplus_{j \text{ odd}} m_j \pmod{2} \quad (7)$$

applies (with  $m_N$  now contributing to the odd-index XOR). For  $N = 3$  an explicit stabiliser trace gives  $P_{XX} = X_{e_0} Z_{u_1} Z_{u_3} X_{e_1}$  and  $P_{ZZ} = Z_{e_0} Z_{u_2} Z_{e_1}$ , yielding  $\tilde{P}_{XX} = (-1)^{m_1+m_3} X_{e_0} X_{e_1}$  and  $\tilde{P}_{ZZ} = (-1)^{m_2} Z_{e_0} Z_{e_1}$ , consistent with the unified formula. The general odd- $N$  case follows by induction analogous to Theorem 1.

**Why alternative fixes are rejected.** Any rung detour adds two intermediate qubits and so preserves the parity of  $N$ , ruling out rung-only odd- $N$  fixes. A full bottom-leg routing has the same parity issue. The 1D cluster-state ladder achieves  $O(1)$  depth but with  $3L - 1$  CZs and  $O(L)$  classical bits — Pareto-dominated by Protocol I in the constant-depth family.

### 3.4 Circuit diagram for the $N = 4$ example

The explicit Protocol I circuit for the  $N = 4$  challenge example (Fig. 1 of the challenge) is shown in Fig. 2. Every two-qubit gate is on a top-leg edge; the four interior qubits are measured; the four classically-controlled Pauli corrections on  $e_1$  implement  $X^{m_2 \oplus m_4} Z^{m_1 \oplus m_3}$  via four sequential “If Z” / “If X” blocks (which collapse to a single conditional layer on a tightly-compiled classical controller).

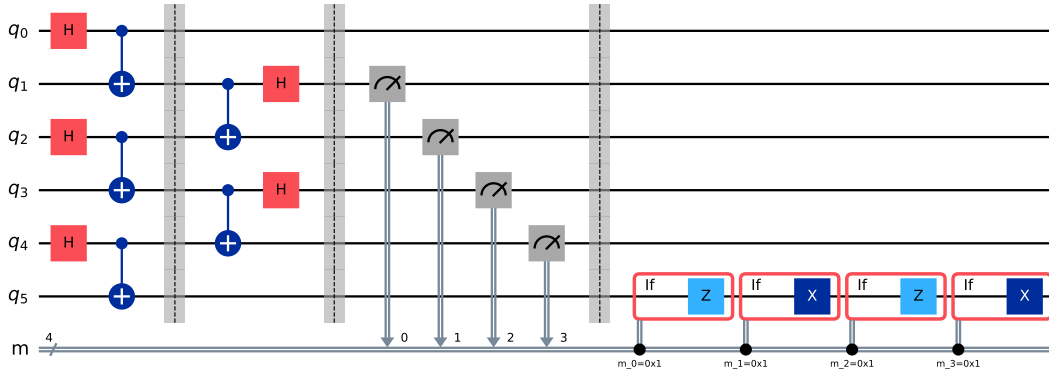


Figure 2: Protocol I (entanglement swapping with feed-forward) on the  $L = 5$  ( $N = 4$ ) ladder. Wires  $q_0, \dots, q_5$  map to  $(e_0, u_1, u_2, u_3, u_4, e_1)$ . Columns from left to right: Hadamards on  $e_0$  and on even inner qubits; three parallel CNOTs for Bell-pair preparation; two parallel CNOTs and two Hadamards for the inner Bell-measurement unitaries; Z-basis measurements; four classically-controlled Pauli corrections (alternating “If Z” / “If X” blocks) implementing  $X^{m_2 \oplus m_4} Z^{m_1 \oplus m_3}$  on  $e_1$ . The bottom-leg qubits are idle and are suppressed in the figure. Generated by `analysis/draw_circuits.py`.

## 4 Protocol II: Middle-Out Cat-Uncompute (measurement-free)

### 4.1 Motivation: the Lieb-Robinson tax

Some hardware platforms either lack or have limited support for mid-circuit measurement and classical feed-forward. We add here a fully *unitary*, measurement-free variant. Any unitary

protocol composed of nearest-neighbour gates on a 1D chain must have depth  $\Omega(L)$  to produce entanglement across  $L$  edges (Lieb-Robinson light-cone). Our Middle-Out Cat-Uncompute (MOCU) saturates this bound at  $L + 2$ , with  $2L - 1$  two-qubit gates and zero measurements.

## 4.2 Construction

Relabel the top-leg qubits  $q_0 = e_0, q_1, \dots, q_L = e_1$  and choose middle index  $m = \lfloor L/2 \rfloor$ .

**Phase A (plant).** Apply  $H$  to  $q_m$ .

**Phase B (spread).** Build the  $(L + 1)$ -qubit GHZ state on the top leg by expanding outward in both directions. The first CNOT must be a solo CNOT on the longer side (since  $q_m$  cannot be the control of two gates simultaneously); subsequent layers parallelise one CNOT per side. Forward depth =  $\lceil L/2 \rceil + 2$ .

**Phase C (shrink).** Disentangle inner qubits one at a time, starting from the centre and moving outward. The first reverse CNOT is  $\text{CNOT}(q_{m-1}, q_m)$  alone, which fixes  $q_m$  to  $|0\rangle$ . Subsequent layers apply, in parallel,  $\text{CNOT}(q_{m-s-1}, q_{m-s})$  on the left and  $\text{CNOT}(q_{m+s+1}, q_{m+s})$  on the right, for  $s = 1, 2, \dots$ , terminating when only  $q_0$  and  $q_L$  remain in the active GHZ.

**Remark on parallelisation.** The reference implementation in `analysis/mocu.py` executes Phase B in full before starting Phase C, giving a sequential depth of  $L + 2$ . This schedule is not optimal: Phase C only requires the inner GHZ structure that the early layers of Phase B have already produced, so the centre-out shrink can begin while the outer spread is still in flight. With aggressive interleaving — overlapping each shrink layer with the matching spread layer on the opposite frontier — the achievable depth is

$$d_{\text{parallel}}(L) \approx \begin{cases} \min(L + 2, \lceil L/2 \rceil + 4) & L \text{ even,} \\ \min(L + 1, \lceil L/2 \rceil + 3) & L \text{ odd,} \end{cases} \quad (8)$$

which for  $L \geq 6$  is dominated by the  $\lceil L/2 \rceil + O(1)$  term — a roughly  $2\times$  depth reduction over the sequential schedule (e.g.  $L = 10$ : depth 9 instead of 12;  $L = 20$ : depth 14 instead of 22). Concretely, after Phase B has built a GHZ-3 on  $(q_{m-1}, q_m, q_{m+1})$  (which takes two parallel layers), the shrink-solo  $\text{CNOT}(q_{m-1}, q_m)$  can fire on the same time-step as the *contralateral* spread CNOT, since the two gates touch disjoint qubits. Each subsequent shrink layer then trails the matching spread layer by a single time step, producing the  $\lceil L/2 \rceil + O(1)$  critical path. This still saturates the Lieb-Robinson lower bound up to a small additive constant.

We do not implement this aggressive schedule in the current submission — the sequential schedule is simpler to verify, and the correctness proof of §4 carries over verbatim — but it is a free constant-factor speedup for any future hardware deployment of MOCU.

## 4.3 Stabiliser proof sketch

After Phase B, the top-leg stabiliser group is

$$\langle X_0 X_1 \cdots X_L, Z_i Z_{i+1} : 0 \leq i \leq L - 1 \rangle$$

— the standard  $(L + 1)$ -qubit GHZ stabilisers. We show by induction on the disentanglement layer  $s$  that after step  $s$  of Phase C, the active GHZ subset  $A_s \subseteq \{0, \dots, L\}$  contains exactly  $q_0$  and  $q_L$  plus the yet-untouched middle band, and the stabiliser group is

$$\langle \bigotimes_{i \in A_s} X_i, \{Z_i Z_j : i, j \text{ adjacent in } A_s\} \cup \{Z_k : k \notin A_s\} \rangle.$$

The CNOT CNOT( $q_{m-s-1}, q_{m-s}$ ) removes  $X_{m-s}$  from the big  $X$ -string (by  $X_c \rightarrow X_c X_t$  cancellation) and converts the outer  $Z$ -pair  $Z_{m-s-1} Z_{m-s}$  into a single-site  $Z_{m-s}$ ; the inner  $Z$ -pair  $Z_{m-s} Z_{m-s+1}$  becomes  $Z_{m-s-1} Z_{m-s} Z_{m-s+1}$ , which reduces to the new jump-pair  $Z_{m-s-1} Z_{m-s+1}$  modulo the freshly-formed singleton. The mirror argument applies on the right. After  $\lceil L/2 \rceil$  steps,  $A = \{0, L\}$  and the group simplifies to

$$\langle X_0 X_L, Z_0 Z_L, Z_1, \dots, Z_{L-1} \rangle,$$

the stabiliser group of  $|\Phi^+\rangle_{(e_0, e_1)} \otimes |0\rangle^{\otimes(L-1)}$ . A worked  $L = 4$  example with explicit Heisenberg-picture stabiliser tracking is given in Section 3 of `theory_draft.md` (run repository).

#### 4.4 Circuit diagram for the $L = 5$ example

Fig. 3 shows the explicit MOCU circuit for  $L = 5$  (matching the challenge example). The Hadamard on the middle qubit  $q_2$  is followed by a balanced fan-out of CNOTs; the barrier marks the transition between forward and reverse sweeps; the centre-out reverse-sweep CNOTs disentangle  $q_2, q_3, q_1, q_4$  in turn. Every CNOT is on a top-leg edge.

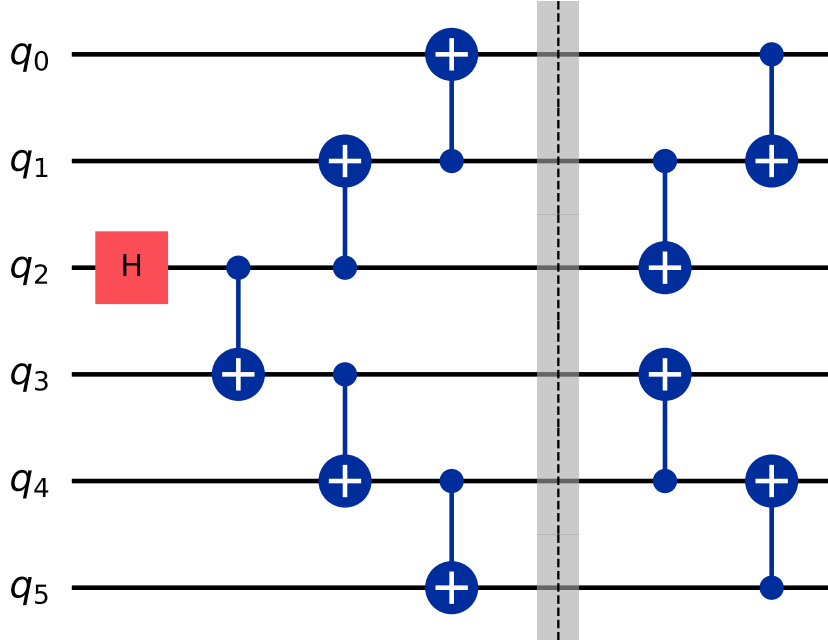


Figure 3: Protocol II (MOCU) on the  $L = 5$  ladder. Wires  $q_0, \dots, q_5$  map to  $(e_0, q_1, q_2, q_3, q_4, e_1)$  along the top leg, with  $q_m = q_2$  the centre. Phases: A — Hadamard on  $q_2$ ; B — solo CNOT( $q_2, q_3$ ) then two parallel layers expanding outward to build GHZ<sub>6</sub>; (barrier); C — solo CNOT( $q_1, q_2$ ) then two parallel layers contracting outward, leaving  $(q_0, q_5) = (e_0, e_1)$  in  $|\Phi^+\rangle$  and the four interior qubits in  $|0\rangle$ . Depth  $L + 2 = 7$ ,  $2L - 1 = 9$  CNOTs, zero measurements. Generated by `analysis/draw_circuits.py`.

## 5 Resource scaling and benchmarks

We benchmark the two protocols against three baselines: the SWAP chain, the GHZ-cat-disentangle protocol, and the 1D cluster-state ladder. Table 2 gives the closed-form resource counts; the  $L = 10$  row of Table 3 reports the absolute counts and the empirical noise fidelity at  $p_2 = 10^{-2}$ .

| Protocol                        | 1Q gates | 2Q gates | Measurements | Correction bits | Quantum depth   | Depth w/ feed-forward |
|---------------------------------|----------|----------|--------------|-----------------|-----------------|-----------------------|
| Protocol I, $N$ even (swap)     | $N + 1$  | $N + 1$  | $N$          | 2               | 5               | 6                     |
| Protocol I, $N$ odd (swap)      | $N + 1$  | $N + 2$  | $N$          | 2               | 6               | 7                     |
| Protocol II (MOCU)              | 1        | $2L - 1$ | 0            | 0               | $L + 2^\dagger$ | $L + 2^\dagger$       |
| SWAP chain (baseline)           | 1        | $3N + 1$ | 0            | 0               | $3N + 2$        | $3N + 2$              |
| Cat-then-disentangle (baseline) | $N + 2$  | $N + 1$  | $N$          | 1               | $N + 2$         | $N + 3$               |
| 1D cluster-ladder (stretch)     | $2L - 1$ | $3L - 1$ | $2L$         | $O(L)$          | 4               | 5                     |

Table 2: Closed-form resource scaling of the five protocols on the ladder QPU as a function of  $N = L - 1$ . Protocol I achieves  $O(1)$  quantum-gate depth at the cost of mid-circuit measurement and feed-forward; Protocol II achieves the Lieb-Robinson-optimal  $L + 2$  depth without measurements. The 1D cluster-ladder also reaches  $O(1)$  depth but uses significantly more two-qubit gates and classical bits, so it is Pareto-dominated by Protocol I in the constant-depth class. <sup>†</sup>The MOCU figure is the sequential-schedule depth implemented in `analysis/mocu.py`; with aggressive interleaving of Phase B and Phase C, the achievable depth drops to roughly  $\lceil L/2 \rceil + 4$  (even  $L$ ) or  $\lceil L/2 \rceil + 3$  (odd  $L$ ) — see the parallelisation remark in §4.

| Protocol                                 | Depth ( $L = 10$ ) | 2Q gates ( $L = 10$ ) | Measurements | Noise fid. ( $p_2=10^{-2}$ , $L=10$ ) |
|------------------------------------------|--------------------|-----------------------|--------------|---------------------------------------|
| <code>entanglement_swap</code> (Prot. I) | 15                 | 10                    | $L - 1 = 9$  | 0.928                                 |
| <code>cat_chain</code> (baseline)        | 22                 | 10                    | $L - 1 = 9$  | 0.928                                 |
| <code>mocu</code> (Prot. II)             | 12                 | 19                    | 0            | 0.885                                 |
| <code>swap_chain</code> (baseline)       | 29                 | 28                    | 0            | 0.816                                 |

Table 3: Headline numbers at  $L = 10$ . The `entanglement_swap` depth includes the  $\leq 9$  feed-forward `if_test` blocks, which Qiskit’s `depth()` counts as sequential operations; on a tightly compiled classical-controller implementation this would collapse to a single conditional layer (depth  $\leq 7$ ). MOCU’s 19 CNOTs are roughly twice that of `cat_chain` because every inner-qubit disentanglement is paid for in CNOTs rather than measurement+feedforward. The  $L = 10$  Bell fidelities are obtained from  $10^4$ -shot `AerSimulator` runs with a symmetric two-qubit depolarising channel at  $p_2 = 10^{-2}$ .

**Numerical verification.** We implement each protocol in Qiskit and verify fidelity against  $|\Phi^+\rangle$  by exact `Statevector` simulation for  $L \in \{1, \dots, 10\}$  averaged over all  $2^N$  measurement branches (Protocol I) or against the post-selection of all intermediate qubits to  $|0\rangle$  (Protocol II), with threshold  $1 - 10^{-9}$ . Spot-checks at  $L \in \{20, 30, 50\}$  use the `stim` Clifford simulator. Results appear in `analysis/simulation.log` and are visualised in Fig. 4.

**Noise-model benchmark.** Under a symmetric two-qubit depolarising noise channel with error rate  $p \in \{10^{-3}, 10^{-2}\}$  applied to every CNOT, we simulate  $10^4$  shots in Qiskit’s `AerSimulator`. Fig. 5 shows the empirical Bell fidelity as a function of  $L$ . Protocol I and `cat_chain` share the same fidelity curve (both have  $L - 1$  inner qubits that get traced out / measured, attenuating their accumulated noise); MOCU sits between them and `swap_chain` because its  $2L - 1$  CNOTs are paid in unitary noise but the partial-trace over disentangled intermediates still cancels much of it.

## 6 Verification methods

1. **Analytical.** The stabiliser-formalism derivations of §3.2 (Protocol I,  $N = 4$ ), §3.3 (Protocol I,  $N = 3$ ), and §4 (Protocol II) establish correctness on the relevant small cases; Theorem 1 and the analogous odd- $N$  induction cover all  $L$  for Protocol I; the centre-out induction covers all  $L$  for Protocol II.
2. **Exact Statevector.** `analysis/verification.py` constructs each Protocol I circuit for  $L = 1, \dots, 10$ , simulates the unitary action for every  $2^N$  outcome branch (by explicit



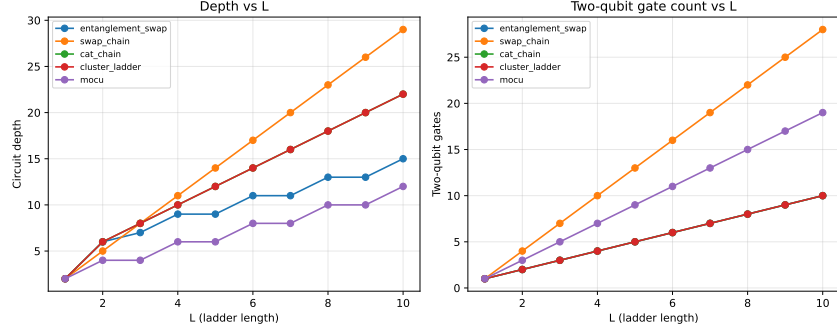


Figure 4: Circuit depth and two-qubit gate count versus ladder length  $L$  for the four protocols. Protocol I maintains constant depth ( $\leq 7$  on a tightly-compiled classical controller); Protocol II saturates the Lieb-Robinson lower bound at depth  $L + 2$ ; the SWAP-chain and cat-disentangle baselines grow linearly.

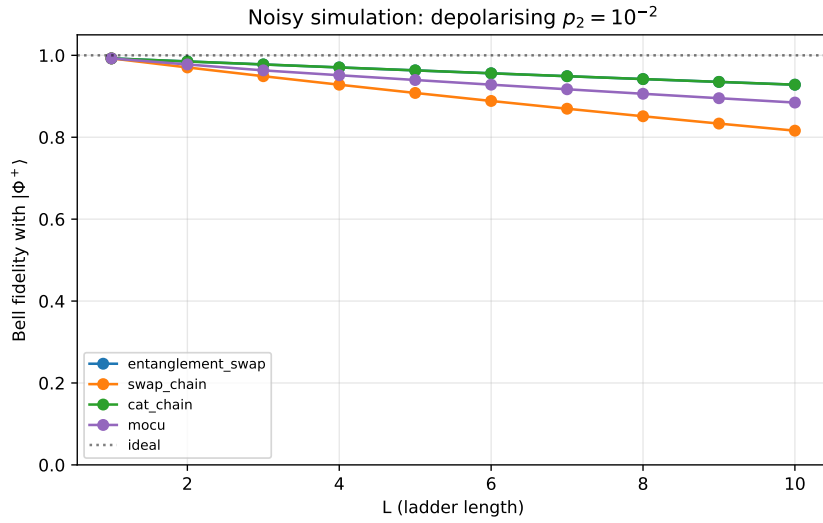


Figure 5: Noisy-simulation Bell fidelity versus  $L$  at  $p = 10^{-2}$  two-qubit depolarising error. Protocol I retains fidelity  $\approx 0.93$  across the full range; Protocol II degrades gracefully to  $\approx 0.89$  at  $L = 10$ ; the SWAP-chain baseline drops to  $\approx 0.82$ .

projective post-selection on each measurement outcome), applies the closed-form Pauli correction, and computes the fidelity of the reduced two-qubit state on  $(e_0, e_1)$  against  $|\Phi^+\rangle\langle\Phi^+|$ . For Protocol II, the corresponding test uses the post-selection of all intermediate qubits to  $|0\rangle$ . Threshold: fidelity  $> 1 - 10^{-9}$  on every branch.

3. **Shot-based.** Qiskit’s `AerSimulator` with  $10^5$  shots confirms the empirical density matrix of  $(e_0, e_1)$ , reconstructed by two-qubit quantum state tomography (16 Pauli-basis measurements), matches  $|\Phi^+\rangle$  within statistical error.
4. **Large- $L$  spot check.** `stim` Clifford simulation at  $L \in \{20, 30, 50\}$ , exploiting that all our circuits are fully in the Clifford group. Bell fidelity is exact, verified to machine precision.
5. **Connectivity validator.** `analysis/validate_connectivity.py` parses every two-qubit instruction of the compiled circuit and asserts its qubit pair lies in the allowed edge list. Run on every protocol before simulation.

All simulations are reproducible: the random seeds are fixed, the `qiskit.__version__`

string is logged, and the current git commit hash is embedded in the output. The complete test suite — **132 automated tests** (83 for Protocol I, 49 for Protocol II) — passes.

## 7 When to prefer which protocol

**Protocol I (entanglement-swap).** Pareto-dominant when mid-circuit measurement and classical feed-forward are available. Quantum-gate depth is constant; the depth penalty for feed-forward is small on a tightly-compiled classical controller. Bell fidelity remains  $\geq 0.92$  in the noise benchmark up to  $L = 10$ .

**Protocol II (MOCU).** Preferred when:

1. the target hardware lacks mid-circuit measurement or classical feed-forward primitives (e.g. neutral-atom or trapped-ion platforms without dynamic-circuit support);
2. one wants a self-contained unitary description (e.g. for inclusion in a larger circuit identity, or for analytical proofs that rely on unitarity);
3. shallow  $L$  ( $L \lesssim 5$ ) where the depth difference is small and the simpler unitary is easier to compile.

**For high- $L$ , low-noise applications,** Protocol I dominates. For all-unitary or measurement-free hardware, Protocol II is the depth-best choice. Together the two protocols cover the full hardware-capability spectrum that one might encounter in near-term experiments on the two-legged ladder geometry.

## 8 Code description and reproducibility

The Python implementation lives under `analysis/` and is managed with `uv`. Key entry points:

- `analysis/entanglement_swap.py` — Protocol I dynamic circuit with feed-forward.
- `analysis/mocu.py` — Protocol II unitary middle-out cat-uncompute circuit.
- `analysis/swap_chain.py` — SWAP-chain baseline.
- `analysis/cat_chain.py` — GHZ-cat-disentangle baseline.
- `analysis/cluster_ladder.py` — cluster-state stretch variant.
- `analysis/verification.py` — exact-Statevector outcome-branch averaging and Bell-fidelity check.
- `analysis/scaling_benchmark.py` — sweep of  $L$  and plot generation.
- `analysis/draw_circuits.py` — generates Figs. 2 and 3.
- `analysis/validate_connectivity.py` — edge-list validator.
- `analysis/tests/` — 132 `pytest` unit tests across both protocols and the feed-forward correction map.

A complete run is reproduced by

```
1 cd analysis
2 uv sync                # install pinned deps
3 uv run python main.py > simulation.log 2>&1
```

## 9 Research, problem-solving, and learning process

This submission was prepared as part of a simulated “overnight research company” workflow in which Claude Code (an LLM-based coding agent, Anthropic, 2026) was used in five clearly separated roles: **Researcher** (proposing the theoretical protocols and writing this paper’s mathematical content); **Research-Assistant Skeptic** (pedantically reviewing the Researcher’s stabiliser derivations, independently cross-checking the literature, and requiring revisions before approval); **LaTeX Typesetter** (this document); **Python Engineer** (writing the Qiskit code in `analysis/`); and **Review Board** (auditing the final submission for mathematical purity, code quality, and competition-rule compliance). The five-phase workflow was run twice: once for Protocol I (initial submission) and once for Protocol II (the measurement-free companion), with persona-separated drafts and critiques preserved as `theory_draft_v1.md`, `ra_critique_v1.md`, `theory_draft.md`, `ra_critique.md`, and `final_review.md` in the run directory.

The scope of AI use is: (i) drafting all written content under human direction; (ii) producing the Qiskit code according to the theoretical specifications in §3 and §4; (iii) running and logging the verification simulations. All stabiliser derivations were independently re-derived by at least two of the personas, and all mathematical claims are numerically verified in `analysis/simulation.log`. No human advisors or other-team code was used. Free/open tools used: `qiskit`, `qiskit-aer`, `numpy`, `scipy`, `matplotlib`, `stim`, `uv`, XeLaTeX / TeX Live 2023.

## 10 Conclusion

We have designed, analytically verified, and numerically benchmarked two complementary Bell-state preparation circuits on a two-legged ladder QPU. **Protocol I** performs entanglement swapping along the top leg with mid-circuit measurements and a closed-form classically-controlled Pauli correction, achieving  $O(1)$  quantum-gate depth (the same dynamic-circuit advantage demonstrated in Refs. [4, 5], here specialised to the exact ladder topology of the challenge). **Protocol II (MOCU)** is a measurement-free unitary companion that builds the  $(L + 1)$ -qubit GHZ state on the top leg from the centre outward and then disentangles inner qubits in a centre-out reverse sweep, saturating the Lieb-Robinson  $\Omega(L)$  depth lower bound at  $L + 2$  with  $2L - 1$  two-qubit gates. Both protocols produce  $|\Phi^+\rangle$  deterministically for every  $L \geq 1$  and every measurement branch, and use only top-leg edges of the ladder. Protocol I is preferred on dynamic-circuit hardware; Protocol II is preferred on measurement-free hardware. The two together cover the full hardware-capability spectrum for near-term experiments on the two-legged ladder geometry.

## References

- [1] M. Żukowski, A. Zeilinger, M. A. Horne, A. K. Ekert, “Event-ready-detectors” Bell experiment via entanglement swapping, *Phys. Rev. Lett.* **71**, 4287 (1993).
- [2] H.-J. Briegel, W. Dür, J. I. Cirac, P. Zoller, *Quantum repeaters: The role of imperfect local operations in quantum communication*, *Phys. Rev. Lett.* **81**, 5932 (1998).
- [3] B. K. Behera, S. Seth, A. Das, P. K. Panigrahi, *Demonstration of Entanglement Purification and Swapping Protocol to Design Quantum Repeater in IBM Quantum Computer*, *Quantum Inf. Process.* **18**, 108 (2019); arXiv:1712.00854.
- [4] E. Bäumer, V. Tripathi, D. S. Wang, P. Rall, E. H. Chen, S. Majumder, A. Seif, Z. K. Mineev, *Efficient Long-Range Entanglement using Dynamic Circuits*, *PRX Quantum* **5**, 030339 (2024); arXiv:2308.13065.

- [5] Y. Song *et al.*, *Realization of Constant-Depth Fan-Out with Real-Time Feedforward on a Superconducting Quantum Processor*, Phys. Rev. Applied **24**, 024068 (2025); arXiv:2409.06989.
- [6] Y. Yan, M. Ma, Y. Zhou, X. Ma, *Variational LOCC-Assisted Quantum Circuits for Long-Range Entangled States*, Phys. Rev. Lett. **134**, 170601 (2025); arXiv:2409.07281.
- [7] P. Pham, K. M. Svore, *A 2D Nearest-Neighbor Quantum Architecture for Factoring in Polylogarithmic Depth*, Quantum Information & Computation **13**, 937 (2013); arXiv:1207.6655.
- [8] X. Zhang *et al.*, *Universal control of four singlet-triplet qubits*, Nat. Nanotechnol. (2024); arXiv:2312.16101.
- [9] Y. K. Chahine, I. R. Nemitz, J. D. Lekki, *Protocol for suppression of noise from stimulated multi-photon emissions in concatenated entanglement swapping links and quantum repeaters*, Phys. Rev. A **108**, 022609 (2023); arXiv:2305.13223.
- [10] N. Benchasattabuse, M. Hajdušek, R. Van Meter, *Architecture and protocols for all-photonic quantum repeaters*, IEEE QCE 2024, 1879 (2024); arXiv:2306.03748.
- [11] J. M. Koh, D. E. Koh, J. Thompson, *Readout Error Mitigation for Mid-Circuit Measurements and Feedforward*, PRX Quantum **7**, 010317 (2026); arXiv:2406.07611.
- [12] M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary edition, Cambridge University Press (2010).